

Week 6: Confidence Intervals

Part 1

Jessie Yeung

STA 220

Winter 2026

Announcements

- Assignment 1 is due this Friday, Feb. 13 at 11:59PM
 - Don't forget to click the submit button at the bottom of the page
 - Late penalty of 20% per day
 - If you cannot complete assignment due to illness or personal emergency, fill out Missed Assignment Form (link in syllabus) within 1 week of the assignment deadline
- Term Test on Tuesday, ^{Feb}~~Oct~~. 24 during class time
 - Official announcement is on Quercus
 - Practice test to be posted
 - No class on Thurs. ^{Feb}~~Oct~~. 26
- Due to reading week, next quiz will be open from Feb. 13 to Feb. 22

Overview

- Over the next few weeks, we will be learning about confidence intervals
- Topics for this week
 - Collecting Data
 - Sampling Distributions as it relates to Inference
 - Confidence Intervals for Proportions

Collecting Data

Populations and Samples

- Before even collecting data, we need to define our **population of interest**.
 - these will be all the people/things that we are trying to learn something about
 - this is not the same thing as all people/things everywhere
 - for example, what if I want to see if a new drug will make a headache go away faster than the old drug.
 - my population of people won't be all people everywhere
 - why would I give the drug to someone that doesn't have a headache...
 - in this case, my population would be all people who have a headache

Alternative to a Census

- A **census** is when you collect information from every person/thing in the population. However, this is usually not feasible.
- When a census is not possible (which is most of the time), then we have to collect information on a smaller portion of the population of interest.
 - this is called the **sample**
 - we still want to talk about the population, but now using the results from the sample
- If we take a census, then we don't have to worry about randomness because we have collected everything.
 - when we take a random sample, now randomness has to be considered.
 - depending on who is in our sample, we will calculate different sample statistics (like sample mean, sample proportion, etc.)

Statistical Inference

- Ideally, we want to take the results/trends from our sample and apply it to the whole population
 - this is called **statistical inference** (infer about the population using results from the sample)
 - we want to determine whether the result we see in our sample is due to chance or is actually the true result in the real world.
- This will require good data collection and a sample that is representative of the population

What are we measuring?

- When we collect a sample, we calculate a **sample statistic** (e.g. sample mean)
- We are attempting to use this statistic to make a guess (or make an inference) about the corresponding value in the population
 - this value in the population is called a **population parameter** (e.g. population mean)
- When we use our sample statistic to make an inference about a population parameter, we call this estimation
 - we calculate a statistic to estimate an unknown parameter
 - this is because we did not take a census, so we do not know this population parameter.

Parameter vs. Statistic

- **Parameters** (or population parameters) are unknown values about the population.
 - They are unknown to us unless we take a census with perfect measurement
 - These are the values that are of interest to us!
- **Statistics** (or sample statistics) are summary measures we can calculate based on our sample data
 - We can calculate them and obtain a number
 - They differ from sample to sample (which is why they have sampling distribution)
- Goal: Use sample statistics to estimate the unknown population parameters
 - Estimate the population mean μ using sample mean $\bar{x}$
 - Estimate the population proportion p using the sample proportion $\hat{p}$

Second half of the course

- We will be working with data!
- In most of the applications in the rest of the course, we will NOT know the true population parameters
- We will be using the data to estimate the true population parameters
- But how do we know if our estimates are good or not?
 - We have to make sure the data is gathered well
- Let's assume the data is gathered perfectly (random sampling with no issues, no missing data, can access whole population...)
 - Is our estimate useful or not? Can use our estimate to say anything about the population parameter?

Recall: Sampling Distributions

Sampling Distributions

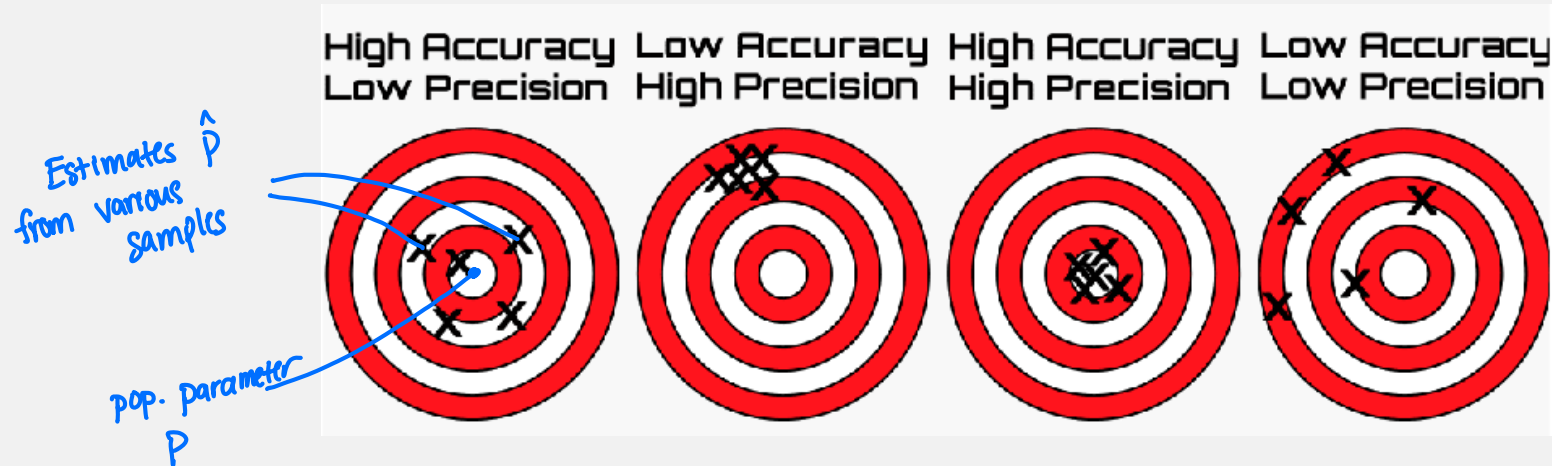
- Let's revisit sampling distributions
- When we study things, we are often dealing with a population that we want to know something about:
 - e.g. suppose we want to know the proportion of Americans that drink more than one soda per day.
 - this proportion is the population parameter p
 - we don't know this value because we can't measure all Americans
 - but we know that the number of Americans that drink more than one soda per day is Binomial, with some probability of success p
 - we want to try to estimate this probability of success with some data

Sampling Distributions

- So we take a sample of say 50 of them. From the sample, we can find the number of them that drink >1 soda/day
 - Suppose we find that 17 of them do... that's a sample proportion of
$$\hat{p} = \frac{17}{50} = 0.34$$
 - This sample proportion is a sample statistic
 - We now want to see if this is a good way to guess (or estimate) this unknown population proportion of all Americans that drink >1 soda/day.
 - In other words, is calculating $\hat{p}$ with a sample of 50 people a good way to guess p ?

What do we mean by a good estimate?

- A good estimate is **high in accuracy**:
 - High accuracy means that it is measuring what we want it to measure and there is no/low bias
- A good estimate is **high in precision**:
 - High precision means that the method of estimating has low variability (estimates are consistent with this method)



Sampling Distributions

- For our particular sample of 50 people, we have $\hat{p} = 0.34$ which is now our estimate for the unknown population parameter p
- We know that the value of $\hat{p}$ will vary from sample to sample.
- But if $\hat{p}$ will vary from sample to sample, how do I know whether the value of $\hat{p}$ from my sample will be a good/bad estimate for p ?
 - Think about the targets. We only get one shot. How do I know which situation I'm in
- The sampling distribution of $\hat{p}$ can tell us about the value of $\hat{p}$ on average and how much its value varies
- Recall that when the CLT applies,

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

Sampling Distributions

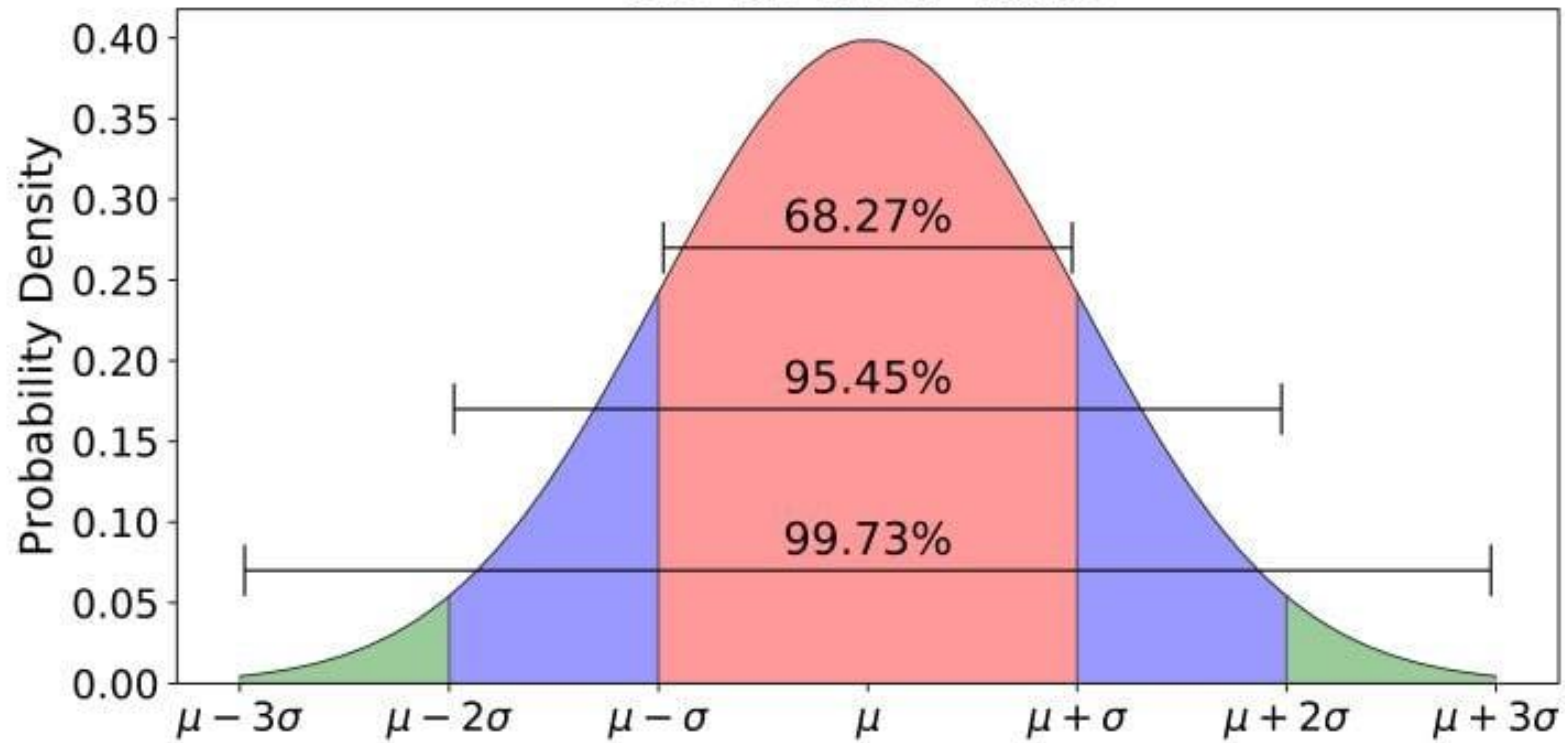
$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

- The **mean** of the sampling distribution tells us that, even though $\hat{p}$ changes from sample to sample, its value on average is p
 - the mean of $\hat{p}$ matches the population proportion, which we are trying to estimate
- The **variance** of the sampling distribution tells us how much $\hat{p}$ changes from sample to sample
 - This gives me some information on how far $\hat{p}$ might be from the unknown p

Aside: 68-95-99 Rule

- The standard deviation, in general, acts as a yard stick
 - it tells us how far things are from the mean/centre
- There is empirical rule to get a rough idea how far we are from the mean when values are normally distributed
 - about 68% of observations are within 1 SD of the mean
 - about 95% are within 2 SD of the mean
 - about 99% are within 3 SD of the mean
- Even if the data is not normal, this can be used as an approximation to tell you how far away you are from the mean
 - Eg. “1 SD away from the mean is not that far, but 4 SD is very far”
 - Use your own judgement on when this approximation is appropriate

68-95-99.7 Rule



Standard Deviation with Normals

- The sampling distribution of $\hat{p}$ tells us that values of $\hat{p}$ are normally distributed
- Due to this, if I have my value of $\hat{p}$ and I know the full distribution of $\hat{p}$, then I can find out how many standard deviations separate $\hat{p}$ and p
 - This would just be a z-score calculation!
- Once I do this, I can find out how likely/unlikely my value of $\hat{p}$ is

Z-score of a Sample Proportion

- Let's take the z-score of our sample proportion, by using its sampling distribution
 - the mean of the sampling distribution is p
 - the standard deviation of the sampling distribution is $\sqrt{\frac{p(1-p)}{n}}$
- To find the z-score, we subtract off the mean, and divide by the standard deviation:
- $Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$
- We know that this process will give us a standard Normal

Soda Example

- So in our example, we have seen that we get a sample proportion of $\hat{p} = 0.34$
- Further suppose the true population proportion is $p = 0.3$
- So, $\hat{p} \sim N\left(0.3, \frac{0.3(1-0.3)}{50}\right)$
- Let's find the z-score for the sample proportion:

$$z\text{-score} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} = \frac{0.34 - 0.3}{\sqrt{\frac{0.3 \times (1-0.3)}{50}}} = 0.62$$

- This seems to suggest that we are reasonably close to the population parameter we want to estimate.
 - but this is still a bit vague... how close is close enough?

Normal Probabilities

- The closer my sample proportion $\hat{p}$ is to the population proportion p , the closer the z-score is to 0.
- My current sample proportion is 0.04 away from the population proportion. What is the probability that a random sample of size 50 will yield a sample proportion that is at most 0.04 away from p ?

$$P(0.26 < \hat{p} < 0.34)$$

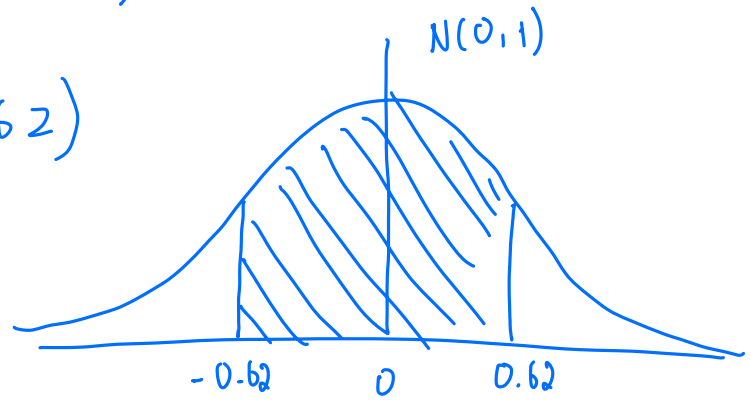
$$= P\left(\frac{0.26 - 0.3}{\sqrt{\frac{0.3 \times 0.7}{50}}} < Z < \frac{0.34 - 0.3}{\sqrt{\frac{0.3 \times 0.7}{50}}}\right)$$

$$= P(-0.62 < Z < 0.62)$$

$$= 1 - 2 \times P(Z < -0.62)$$

$$= 1 - 2 \times 0.2676$$

$$= 0.4648$$



But...Population parameters are unknown

- I can find all this information because we were given the value of $p = 0.3$
- This gave us $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right) \equiv N\left(0.3, \frac{0.3(1-0.3)}{50}\right)$ which tells us everything we need to know about the behaviour of $\hat{p}$
- Recall that in real life we don't know the value of population parameters
 - Or else why even collect data and do statistics???

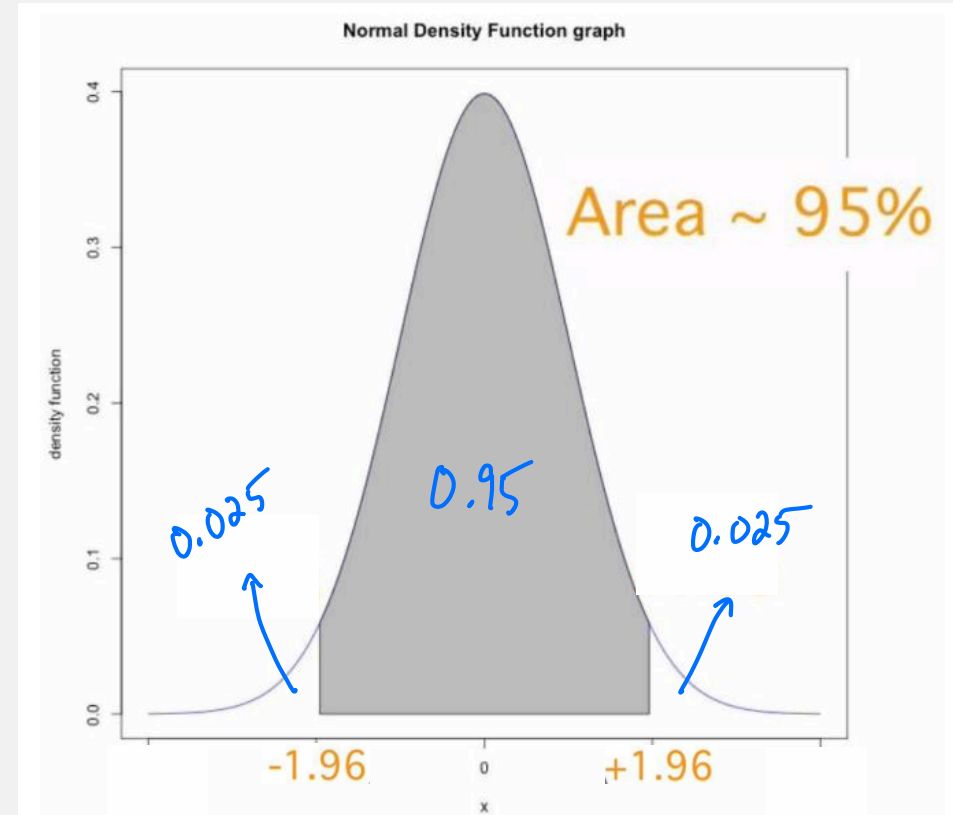
Idea of Confidence Intervals

- Even though I won't be able to fill in the value of p into the sampling distribution, all is not lost
- We can still use the sampling distribution in order to create a **confidence interval**
- Confidence intervals are an interval that describes:
 - It describes a range of plausible values for the population parameter
 - It describes the confidence we have in our estimate
- Statements like “We are 95% confident that between 63% and 69% of all adult Canadians support the government's response to the refugee crisis” is example of a confidence interval

Confidence Intervals for Proportions

Standard Normal Distribution

- The standard Normal distribution is very useful because we can graph it and we can determine the probabilities in different regions.
- For example, the area under the curve between -1.96 and 1.96 is equal to 95% of the total area.
- If a random variable follows a standard Normal distribution only 5% of the time will an observation be less than -1.96 or greater than 1.96 .



Standard Normal Distribution

- In other words,

$$P(-1.96 < Z < 1.96) = 0.95$$

- We also know that

$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

- Let's substitute in the standardized sampling distribution of $\hat{p}$ into the first statement

Deriving the confidence interval

$$P(-1.96 < Z < 1.96) = 0.95$$

Sub in

$$P\left(-1.96 < \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} < 1.96\right) = 0.95$$

multiply by
denominator

$$P\left(-1.96 \sqrt{\frac{p(1-p)}{n}} < \hat{p} - p < 1.96 \sqrt{\frac{p(1-p)}{n}}\right) = 0.95$$

subtract
 $\hat{p}$

$$P\left(-\hat{p} - 1.96 \sqrt{\frac{p(1-p)}{n}} < -p < -\hat{p} + 1.96 \sqrt{\frac{p(1-p)}{n}}\right) = 0.95$$

$2 < 3$
 $-2 > -3$

multiply
by -1

$$P\left(\hat{p} + 1.96 \sqrt{\frac{p(1-p)}{n}} > p > \hat{p} - 1.96 \sqrt{\frac{p(1-p)}{n}}\right) = 0.95$$

Rearrange

$$P\left(\hat{p} - 1.96 \sqrt{\frac{p(1-p)}{n}} < p < \hat{p} + 1.96 \sqrt{\frac{p(1-p)}{n}}\right) = 0.95$$

Therefore, the 95% confidence interval for p is

$$\left(\hat{p} - 1.96 \sqrt{\frac{p(1-p)}{n}}, \hat{p} + 1.96 \sqrt{\frac{p(1-p)}{n}}\right)$$

Standard Error

- The **standard error** refers to the standard deviation of an estimator (sampling statistic)
- It represents how precise an estimate is

- In the case of the sample proportion, $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

- So, another way to state the confidence interval is:

$$\hat{p} \pm 1.96 \times SE(\hat{p})$$

- Or

$$\hat{p} \pm 1.96 \times \sqrt{\frac{p(1-p)}{n}}$$

Other levels of confidence

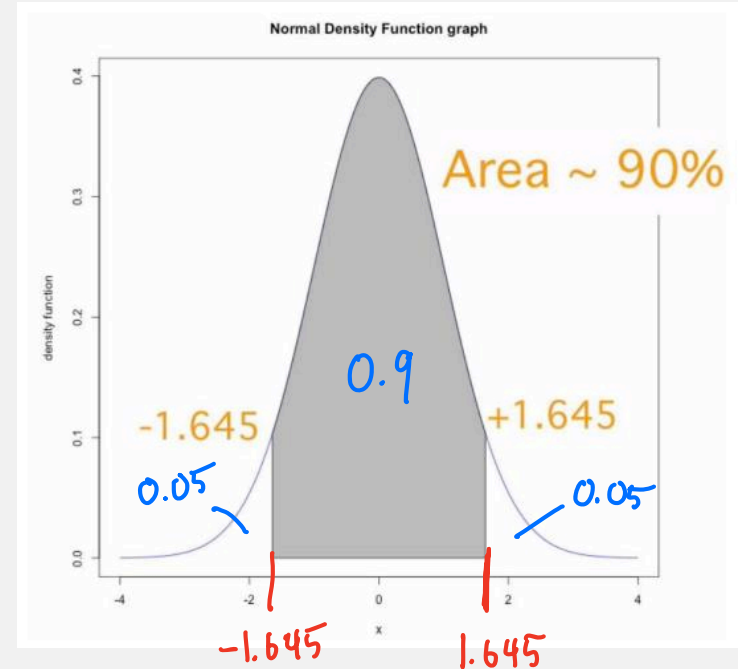
- Not all confidence intervals need to be at the 95% confidence interval. They can be at any level of confidence.
- Suppose you were interested in a 90% confidence interval. How would the formula change?

90% CI:

$$\hat{p} \pm 1.645 \times SE(\hat{p})$$

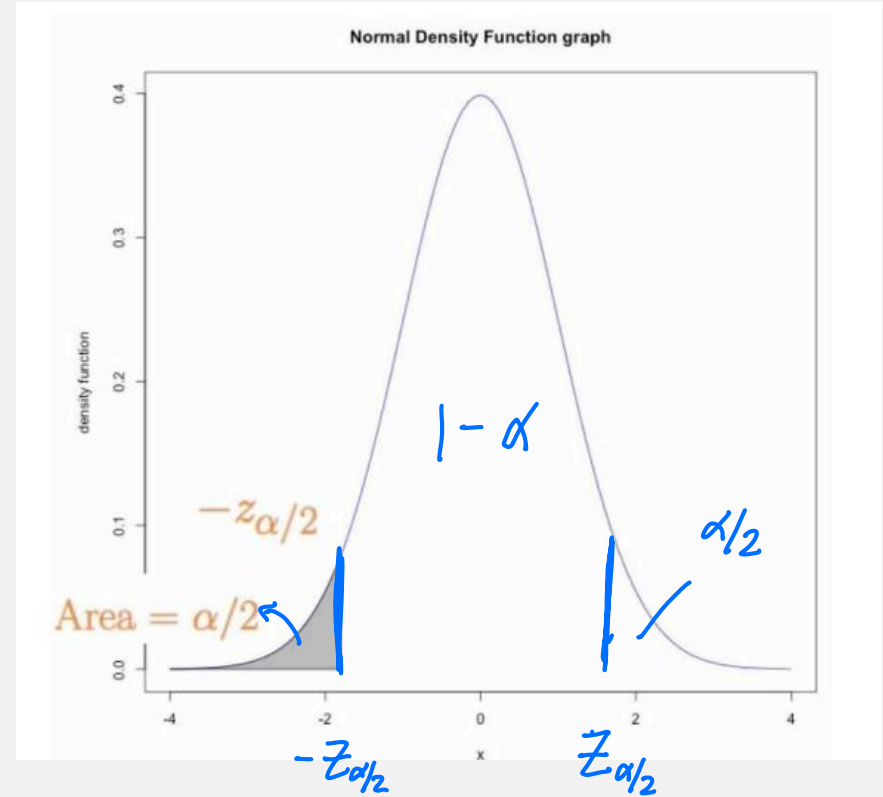
OR

$$\left(\hat{p} - 1.645 \sqrt{\frac{p(1-p)}{n}}, \hat{p} + 1.645 \sqrt{\frac{p(1-p)}{n}} \right)$$



Any level of confidence

- In general, you can find a $(1 - \alpha)\%$ confidence interval for any value α
 - A 95% confidence level corresponds to $\alpha = 0.05$
- In general, for any value α , you can find a value $z_{\alpha/2}$ such that the area under the standard Normal curve which is less than $-z_{\alpha/2}$ is equal to $\alpha/2$.
 - Eg. For $\alpha = 0.05$, $z_{\alpha/2} = 1.96$ since $P(Z < -1.96) = \frac{0.05}{2} = 0.025$
- The value $z_{\alpha/2}$ is called the **critical value**



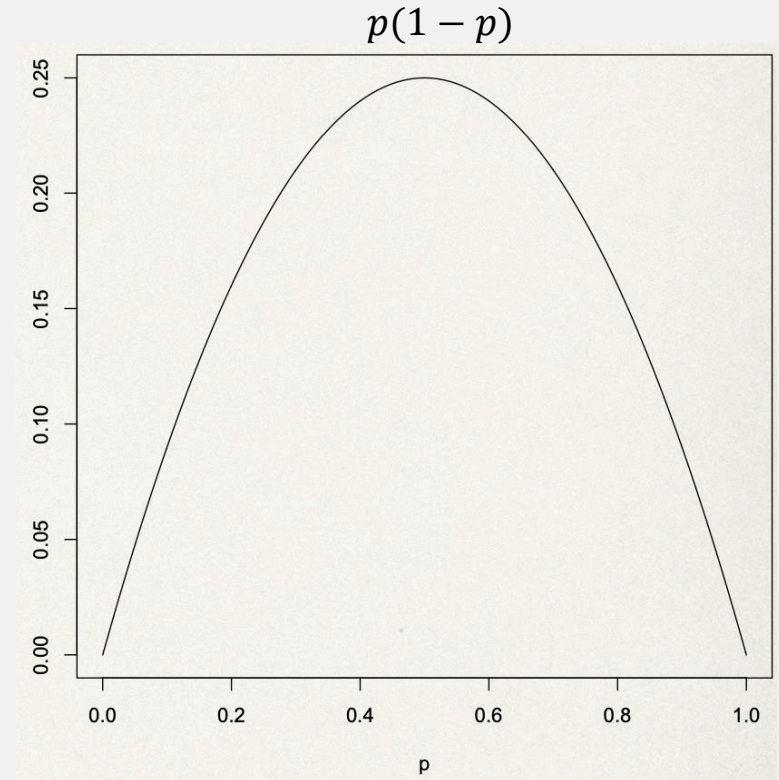
Don't know parameter p ...

- But we don't actually know what the population proportion is
- So we can't use the interval as it is, because the standard deviation of the sampling distribution uses p .
- We have two options for dealing with this:
 1. Use $\hat{p}$ in place of p
 - if our sample is good, then it's reasonable to think that $\hat{p}$ will be close to p
 2. Use $p = 0.5$
 - this is a conservative/safe choice - it will give you the largest possible interval

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

How is $p = 0.5$ the most conservative choice?

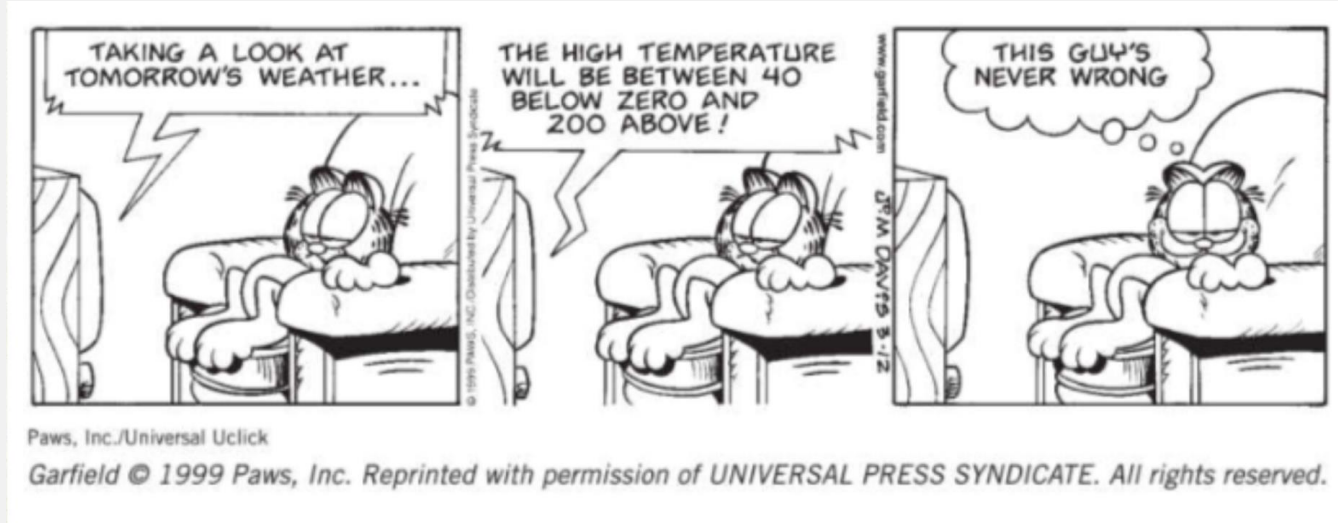
- We can write our interval as
$$\hat{p} \pm z_{\alpha/2} \times SE(\hat{p})$$
- The $SE(\hat{p})$ will impact how wide the interval is
- We see that when $p = 0.5$, the standard error is the largest it can be.
- By making the confidence interval the widest it can be, we make sure that we reach the desired confidence level



Balance between error and confidence

width
↑

- So what happens when we use a higher confidence level?
 - It turns out that this makes my confidence interval bigger too.
- This means we have a trade off between the width of the interval (i.e. precision) and the confidence level.



Example: Soda

We found from a sample of 50 Americans that 34% of them have more than 1 soda per day. Build a 95% confidence interval, using $p \approx \hat{p}$

$$\hat{p} = 0.34$$

$$\hat{p} \pm 1.96 \sqrt{\frac{p(1-p)}{n}}$$

$$n = 50$$

$$\Rightarrow 0.34 \pm 1.96 \sqrt{\frac{0.34(1-0.34)}{50}}$$

$$\Rightarrow (0.209, 0.471)$$

Example: Soda

Build a 95% confidence interval, using $p = 0.5$.

$$\hat{p} \pm 1.96 \sqrt{\frac{p(1-p)}{n}}$$

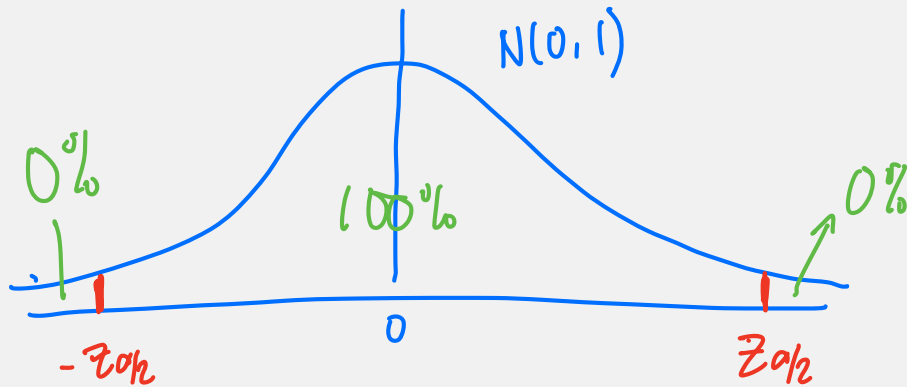
$$\Rightarrow 0.34 \pm 1.96 \sqrt{\frac{0.5(1-0.5)}{50}}$$

$$\Rightarrow (0.201, 0.479)$$

Example: Soda

Build a 100% confidence interval, using $p = 0.5$. What happens?

Let's try to find the critical value



We need a value such that

$$P(Z < -z_{\alpha/2}) = \frac{\alpha}{2} = 0$$

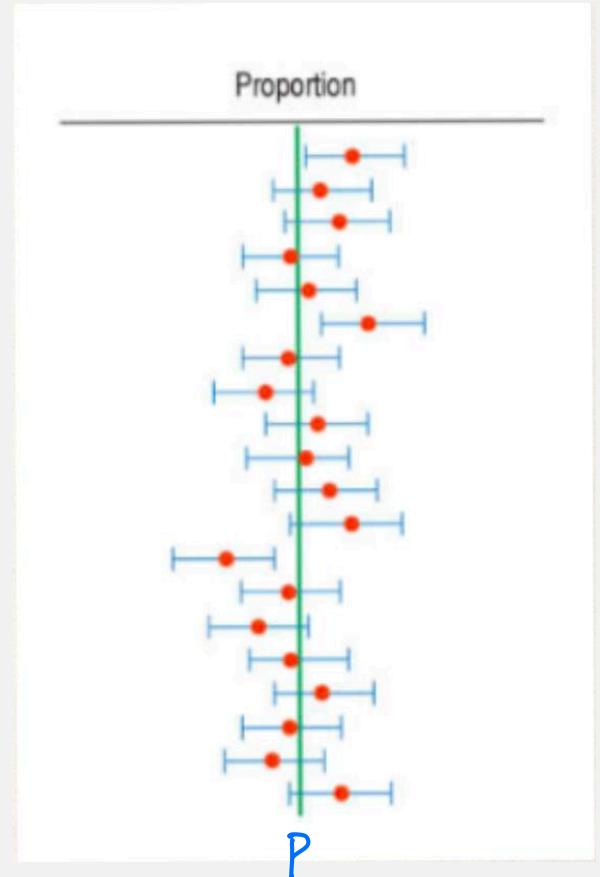
The critical value is infinity.

Therefore, the CI is infinitely wide.

You can use any confidence level you want, but must pick one that is reasonable.

Meaning of Confidence Intervals

- What do these intervals mean?
- They are a statement about our sample.
- When we are talking about a 95% confidence interval, we are saying that, for all samples of the same size as mine, 95% of confidence intervals on those samples will overlap with the population parameter I am trying to estimate.



How to interpret confidence intervals?

- “We are 95% confident that population proportion of _____ is between a and b”
- Avoid talking about probabilities that p is in the interval
 - Is it correct to say that there is a 95% probability that p is between a and b?
 - No, p is a fixed value that is unknown to us. Either p is in the interval or not.
 - Due to this, $P(a < p < b)$ is either 0 or 1
 - Instead, the correct way to think of this is that 95% of the confidence intervals created in this way across all possible samples will contain the true p

$$P(2 < 3) = 1$$

Next Week

- Next week we will see confidence intervals for means